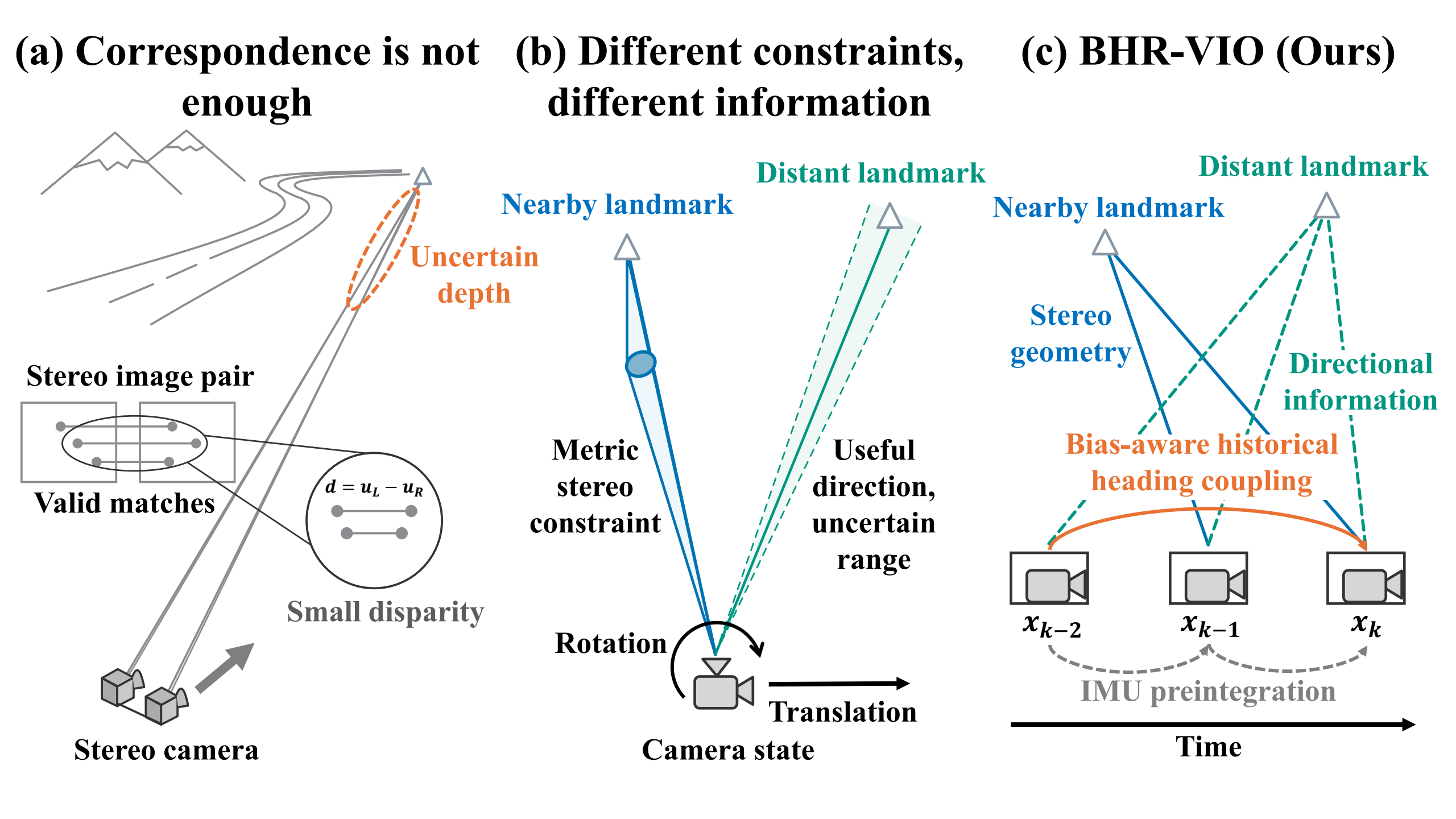

(a) Correspondence is not enough
(b) Different constraints, different information
(c) BHR-VIO (Ours)
Uncertain depth
Stereo image pair
Valid matches
$d = u_L - u_R$
Small disparity
Stereo camera
Nearby landmark
Distant landmark
Metric stereo constraint
Useful direction, uncertain range
Rotation
Translation
Camera state
Nearby landmark
Distant landmark
Stereo geometry
Directional information
Bias-aware historical heading coupling
x_{k-2}
x_{k-1}
x_k
IMU preintegration
Time